

GELOY™ Resin HRA222F - Asia

Polycarbonate + ASA

SABIC

PROSPECTOR®

www.ulprospector.com

Technical Data

Product Description

PC/ASA flame retardant, excellent weatherability.

General

Material Status	• Commercial: Active		
UL Yellow Card ¹	• E207780-100044615 • E207780-100183331		
Search for UL Yellow Card	• SABIC • GELOY™ Resin		
Availability	• Asia Pacific		
Uses	• Appliances • Construction Applications • Electrical Parts • Electrical/Electronic Applications	• Electronic Displays • Lawn and Garden Equipment • Lighting Applications • Medical/Healthcare Applications	• Outdoor Applications • Personal Care • Surgical Instruments • Water Management
Also Available In	• Europe	• Latin America	• North America

Physical	Nominal Value Unit	Test Method
Density	1.17 g/cm ³	ISO 1183
Melt Volume-Flow Rate (MVR) (260°C/2.16 kg)	13 cm ³ /10min	ISO 1133
Molding Shrinkage - Flow ³	0.40 to 0.60 %	Internal Method
Water Absorption		ISO 62
Saturation, 23°C	0.60 %	
Equilibrium, 23°C, 50% RH	0.20 %	
Outdoor Suitability	f1	UL 746C

Mechanical	Nominal Value Unit	Test Method
Tensile Modulus		
-- ⁴	2590 MPa	ASTM D638
--	2520 MPa	ISO 527-1/1
Tensile Strength		
Yield ⁵	63.0 MPa	ASTM D638
Yield ⁶	59.0 MPa	ASTM D638
Yield	62.0 MPa	ISO 527-2/5
Yield	60.0 MPa	ISO 527-2/50
Break ⁶	66.0 MPa	ASTM D638
Break ⁵	56.0 MPa	ASTM D638
Break	51.0 MPa	ISO 527-2/5
Break	45.0 MPa	ISO 527-2/50
Tensile Elongation		
Yield ⁶	4.2 %	ASTM D638
Yield ⁵	4.3 %	ASTM D638
Yield	4.3 %	ISO 527-2/5
Yield	4.4 %	ISO 527-2/50
Break ⁶	> 100 %	ASTM D638
Break ⁵	> 100 %	ASTM D638
Break	> 50 %	ISO 527-2/5 ISO 527-2/50



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Mechanical	Nominal Value Unit	Test Method
Flexural Modulus ⁷	2510 MPa	ISO 178
Flexural Stress ^{7, 8}	93.0 MPa	ISO 178
Impact	Nominal Value Unit	Test Method
Charpy Notched Impact Strength ⁹ (23°C)	15 kJ/m²	ISO 179/1eA
Notched Izod Impact		
0°C	290 J/m	ASTM D256
23°C	390 J/m	ASTM D256
-30°C ¹⁰	9.0 kJ/m²	ISO 180/1A
0°C ¹⁰	12 kJ/m²	ISO 180/1A
23°C ¹⁰	17 kJ/m²	ISO 180/1A
Thermal	Nominal Value Unit	Test Method
Deflection Temperature Under Load ¹¹		
0.45 MPa, Unannealed, 4.00 mm, 100 mm Span	99.0 °C	ISO 75-2/Be
1.8 MPa, Unannealed, 4.00 mm, 100 mm Span	88.0 °C	ISO 75-2/Ae
Vicat Softening Temperature		
--	104 °C	ISO 306/B120
--	102 °C	ISO 306/B50
--	111 °C	ISO 306/A50
Ball Pressure Test (73 to 77°C)	Pass	IEC 60695-10-2
CLTE		ISO 11359-2
Flow : -30 to 80°C	6.9E-5 cm/cm/°C	
Transverse : -30 to 80°C	7.1E-5 cm/cm/°C	
Transverse : 23 to 80°C	7.5E-5 cm/cm/°C	
RTI Elec	90.0 °C	UL 746B
RTI Imp	90.0 °C	UL 746B
RTI Str	90.0 °C	UL 746B
Electrical	Nominal Value Unit	Test Method
Comparative Tracking Index (CTI)	PLC 0	UL 746A
Comparative Tracking Index	600 V	IEC 60112
Flammability	Nominal Value Unit	Test Method
Flame Rating (2.0 mm)	V-0	UL 94
Oxygen Index	29 %	ISO 4589-2
Injection	Nominal Value Unit	
Drying Temperature	80 to 90 °C	
Drying Time	2.0 to 4.0 hr	
Suggested Max Moisture	0.020 %	
Hopper Temperature	60 to 80 °C	
Rear Temperature	200 to 230 °C	
Middle Temperature	220 to 260 °C	
Front Temperature	230 to 270 °C	
Nozzle Temperature	220 to 260 °C	
Processing (Melt) Temp	230 to 270 °C	
Mold Temperature	50 to 70 °C	



Notes

- ¹ A UL Yellow Card contains UL-verified flammability and electrical characteristics. UL Prospector continually works to link Yellow Cards to individual plastic materials in Prospector, however this list may not include all of the appropriate links. It is important that you verify the association between these Yellow Cards and the plastic material found in Prospector. For a complete listing of Yellow Cards, visit the UL Yellow Card Search.
- ² Typical properties: these are not to be construed as specifications.
- ³ Tensile Bar
- ⁴ 5.0 mm/min
- ⁵ Type I, 50 mm/min
- ⁶ Type I, 5.0 mm/min
- ⁷ 2.0 mm/min
- ⁸ at Yield
- ⁹ 80*10*4 sp=62mm
- ¹⁰ 80*10*4 mm
- ¹¹ 120*10*4 mm



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Where to Buy

Supplier

SABIC

Web: <http://www.sabic.com/>

Distributor

Please contact the supplier to find a distributor for GELOY™ Resin HRA222F - Asia

